

How to present: ARM-450 rev I.1 kinematics

One slide per idea; about 1 minute each, 15-18 minutes in all. For each slide: what to SAY, and short answers IF ASKED.

Tips:

1. Start from the question: what can this arm reach and draw, and how do we compute it?
2. Say every number with its check ("FK matches the CAD to $1e-13$ mm").
3. Linger on the task-workspace slide: 172.5 degrees is the result the guide will remember.
4. Show the GIF on the CAD-drawing slide in slideshow mode.
5. If you get lost: convention -> table -> FK -> IK -> workspace -> drawing.

Slide 1 -- Title

ARM-450 rev I.1 -- kinematics

DH table, forward and inverse kinematics, Jacobian, workspace -- in my convention;
Modified (Craig) DH + velocity-propagation Jacobian (after the myCobot 280 thesis and Rsine)
Geometry from the rev I.1 CAD: d1 = 90, a2 = 119, d4 = 221.369, d_T = 88.089 mm; spherical wrist
FK = CAD to 2.98e-13 mm, = URDF to 2.55e-13 mm (2000 poses)
Closed-form IK: 8 branches, exact, 0.2 ms; numerical solvers compared
Pen straight down impossible (172.5 deg of pitch) -> tilted pen on the table; vertical board works

SAY

This is the kinematics of the arm I designed and printed, done the way I did the myCobot thesis: Modified DH, velocity-propagation Jacobian, DLS checked with ikpy. Everything is measured on the CAD, and verified.

IF ASKED

Why not the myCobot? -- Same method, but on my own hardware: the numbers now drive a real design.

Slide 2 -- My convention

My convention: Modified (Craig) DH

1 The convention: Modified (Craig) DH
Frame $i-1$ is attached to link i . Frames use only R and T . R = along joint axis (right-hand rule, + if up) of frame -- from the vertical joint $(i, 0, 0)$ point to the joint $(i, 0, 0)$ point + θ_i as a rotation angle from the joint forward.
Frame i is attached to link $i+1$. $T_{i-1,i}$ along the common normal from Z_{i-1} to Z_i . $T_{i-1,i}$ completes the right-handed set.

a_{i-1} link length: the mutual perpendicular distance between Z_{i-1} and Z_i , measured along X_{i-1} .

α_{i-1} link twist: the angle from Z_{i-1} to Z_i , measured about X_{i-1} .

d_i link offset: the distance from X_{i-1} to X_i , measured along Z_i . (kinematic variable)

θ_i joint angle: the angle from X_{i-1} to X_i , measured about Z_i . (revolute variable)

$${}^{i-1}T_i = Rot_{X_{i-1}}(a_{i-1}) Trans_{Z_i}(d_i) Rot_{Z_i}(\theta_i) Trans_{X_i}(0)$$

$${}^{i-1}T_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & c\alpha_{i-1} d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & s\alpha_{i-1} d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

unofficial credit: per Craig, ITC is exactly this matrix; the analysis derives the table from the DH with those rules (unofficial, not from Craig)

SAY

Frame i is on link i . a and α are measured along and about X of the previous frame, d and θ along and about Z of this frame. The transform is Rot_x , $Trans_x$, Rot_z , $Trans_z$.

IF ASKED

Standard vs modified DH? -- Standard puts frame i at the far end of link i and uses a_i , α_i ; modified (Craig) attaches it at the near end, which makes the velocity propagation clean.

Slide 3 -- DH table

DH table (derived from the CAD servo axes)

ARM-450 rev 1.1 -- Modified (Craig) DH parameters

i	a_{i-1} (mm)	a_i (mm)	θ_i (deg)	θ_i from servo (i)	servo range	joint
0	0	0.000	0.000	$\theta_0 = q_0$	$q_0 \pm 90^\circ$	(0) base roll
1	80	0.000	0.000	$\theta_1 = q_1 + 90^\circ$	$q_1 \pm 90^\circ$	(1) shoulder
2	0	119.000	0.000	$\theta_2 = q_2 + 90^\circ$	$q_2 \pm 172^\circ$	(2) elbow
3	90	0.000	121.000	$\theta_3 = q_3$	$q_3 \pm 90^\circ$	(3) forearm roll
4	80	0.000	0.000	$\theta_4 = q_4$	$q_4 \pm 90.3^\circ$	(4) wrist pitch
5	90	0.000	0.000	$\theta_5 = q_5$	$q_5 \pm 90^\circ$	(5) wrist roll

$T_0^1 = Rot_z(a_{0-1}) Trans_x(a_0) Rot_z(\theta_0) Trans_x(d_1)$ $T_1^2 = Trans_x(d_2)$ $d_0 = 88.089$ mm

$T^6 = T_0^1 T_1^2 T_2^3 T_3^4 T_4^5 T_5^6$ home: $T^6 = Trans_z(518.458)$

Derived from the CAD servo axes and checked: every link transform has the exact Craig form, the frames match the model's to 1e-04 mm. Separates wrist pitch, roll, roll at 90, 450, 900 mm above the base. -> closed form IK.

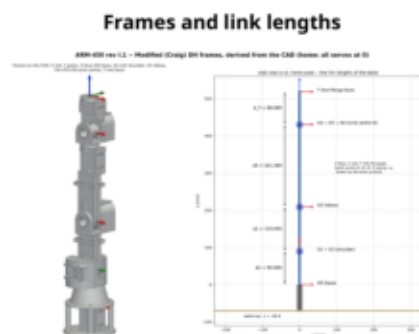
SAY

I did not type these numbers: the script builds the frames from the measured servo axes with my rules and reads the table back; every link transform has the exact Craig form. d_4 is 221.369: elbow to wrist centre.

IF ASKED

Why are there -90 and +90 offsets? -- So that theta is zero at the natural DH position while the servos read 0 with the arm straight up. Why minus signs on some servos? -- four servos are mounted turning about -z or -y.

Slide 4 -- Frames and link lengths



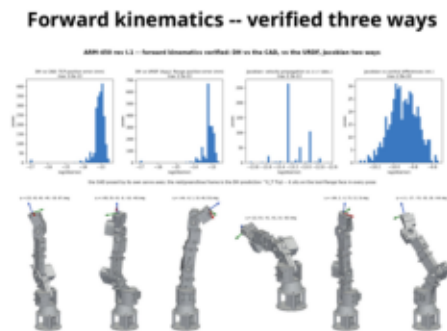
SAY

The wrist centre is where J4, J5 and J6 meet -- that is what makes the IK closed-form.

IF ASKED

What is the tool point? -- The centre of the tool-flange face, 88.089 mm beyond the wrist centre.

Slide 5 -- FK verified



SAY

Three independent checks: the CAD itself moved by its own servo axes (no DH at all), the URDF in RViz through ikpy, and the Jacobian two ways. Agreement is 2.98e-13 mm.

IF ASKED

How do you know the CAD check is independent? -- It uses the product of exponentials of the measured screw axes; the DH table is never read.

Slide 6 -- Jacobian by velocity propagation

Jacobian by velocity propagation

4. Jacobian by velocity propagation (Craig)

$${}^{i-1}\dot{\mathbf{x}}_{i+1} = {}^{i-1}\mathbf{J}_i \dot{\mathbf{q}}_i + \dot{\mathbf{x}}_{i+1}$$

$${}^{i-1}\dot{\mathbf{x}}_{i+1} = {}^{i-1}\mathbf{J}_i \dot{\mathbf{q}}_i + \dot{\mathbf{x}}_{i+1}$$

revolute joint: drop the $\dot{\mathbf{q}}_i$ term; prismatic: drop the $\dot{\mathbf{q}}_i$ term (kept in the code)

tool face: $\dot{\mathbf{x}}_{i+1} = \dot{\mathbf{x}}_{i+1} + \dot{\mathbf{x}}_{i+1} \times \mathbf{z}_i \cdot \dot{\mathbf{q}}_i$

base frame: $\dot{\mathbf{x}}_1 = \mathbf{J}_1 \dot{\mathbf{q}}_1$; $\dot{\mathbf{x}}_{i+1} = \mathbf{J}_i \dot{\mathbf{q}}_i$; column i of $\mathbf{J}$ = the tip twist for $\dot{\mathbf{q}}_i = 1$

Implemented in armSimulator.py (appendix): each column is carried from frame to frame from the base to the tip exactly as above, then summed in the base frame. Checked against the geometric form $\mathbf{J}_i = \mathbf{J}_{i-1} \mathbf{J}_i$ and against central differences of the FK.

FORWARD KINEMATICS = JACOBIAN PROPAGATION (2008 version prints inside the joint twists); Jacobian (2010)
 velocity propagation 1 vs 2 x 10⁻¹⁰ s = 0.11
 velocity propagation 1 vs 2 using 1000 random vectors
 manipulability (Hollerbach): $w = \sqrt{\det(\mathbf{J}_i^T \mathbf{J}_i)}$; condition number $\kappa = \sigma_{\max}/\sigma_{\min}$

SAY

Each column is the tip twist for a unit rate of one joint, carried from the base to the tip with the two propagation equations, then rotated into the base frame.

IF ASKED

Why not just differentiate FK? -- Finite differences are only approximate and break down near singularities; propagation is exact. They agree to 2.91e-10 (relative).

Slide 7 -- IK in closed form

Inverse kinematics in closed form

5 Inverse kinematics in closed form (spherical wrist, Pieper)

1. initial condition: $\vec{W} = \vec{p} - \vec{q} \cdot \vec{p} / |\vec{p}|^2$ (4.25), $\vec{W}$ is fixed at W_1
2. base: $\vec{U}_1 = \text{atan2}(W_1, W_2) \cdot \vec{e}_1 + \vec{e}_2$ (see reaction track)
3. zero plane: $\vec{r} = \vec{q}_1 W_2 - \vec{q}_2 W_1$, $\vec{r} = W_1 \cdot \vec{e}_1 - \frac{r^2}{2|\vec{q}_1|^2} \frac{\vec{q}_1}{|\vec{q}_1|}$
 $\vec{U}_2 = \text{atan2}(W_2, W_1) \cdot \vec{e}_1 - \text{atan2}(2|\vec{q}_1| W_1, \vec{r}) \cdot \vec{e}_2 + \text{atan2}(\vec{p}, \vec{r})$
 $\vec{U}_3 = \vec{q}_1 \cdot 90^\circ$, $\vec{U}_4 = \vec{q}_2 \cdot 90^\circ$ (p = beam from vertical)
4. zero: $\text{atan2}(W_2 / W_1^2, W_1^2) = \text{atan2}(W_2, W_1) = \vec{U}_1$ (4.26), $\vec{U}_1 = M \cdot \vec{U} \cdot \vec{U} / 2$, zero
 $\vec{U}_1 = \text{atan2}(2|\vec{q}_1| W_1, W_1^2) \cdot \vec{e}_1 + \text{atan2}(2|\vec{q}_2| W_2, W_2^2) \cdot \vec{e}_2$, $\vec{U}_2 = \text{atan2}(2|\vec{q}_1| W_1, W_1^2)$, $\vec{U}_3 = \text{atan2}(2|\vec{q}_2| W_2, W_2^2)$
5. 2 (Schubert) + 2 (Lefebvre) + 2 (Jardot) = up to 8 solutions, the point links being the valid ones

[illegible]

SAY

Because of the spherical wrist, position and orientation separate: back off the tool to the wrist centre, solve a 2-link planar arm, then read the wrist angles as Z-Y-Z Euler angles.

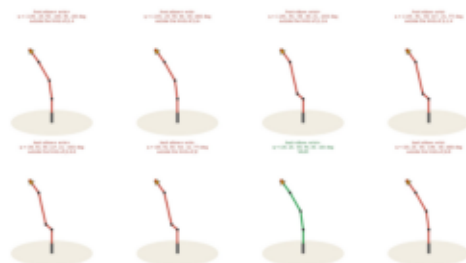
IF ASKED

How many solutions? -- Up to 8; the joint limits leave 1 or 2 (mean 1.37).

Slide 8 -- The 8 branches

The 8 branches of one pose

(Closed from 10:00 to 11:00 a.m. for one hour prior to 10:00 a.m. to 11:00 a.m. – green + inside every label field)



SAY

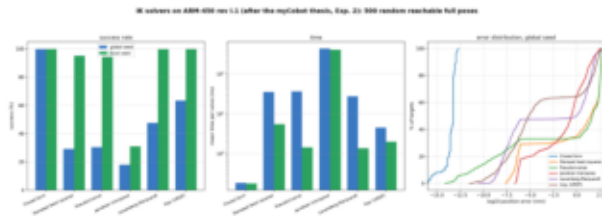
Here are all eight for one pose; only the green one is inside every joint limit.

IF ASKED

Which do you use on the robot? -- The valid one nearest the current joints, so the arm never jumps.

Slide 9 -- IK solvers compared

IK solvers compared (thesis Exp. 2)



SAY

Like thesis experiment 2. The closed form is exact and fast. The iterative solvers are local: near the answer they are fine, from far away they fail often.

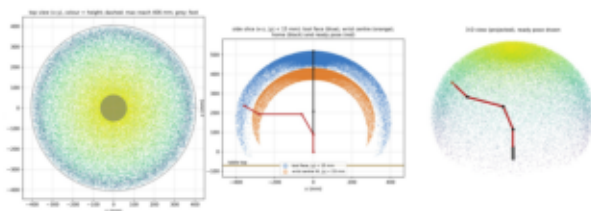
IF ASKED

Then why keep DLS? -- It is general (works without a spherical wrist) and is what follows a path; the closed form seeds it.

Slide 10 -- Reachable workspace

Reachable workspace

ARM-450 rev 1.1 -- reachable workspace (front flange face, 400000 random poses, every joint inside its limits)



SAY

Monte Carlo over the joint limits: reach 405.5 mm from the J1 axis.

IF ASKED

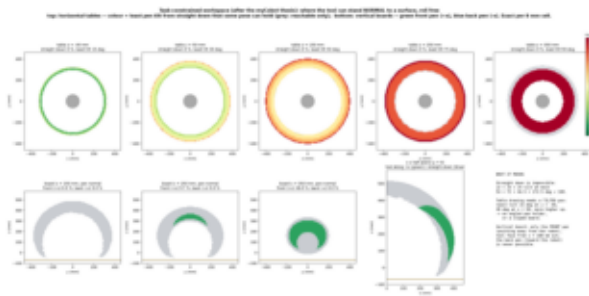
Why the hole above the base? -- The pitch joints are limited, so the arm cannot fold back on itself.

Slide 11 -- Task-constrained workspace

SAY

The key finding: the pen can NEVER point straight down -- the pitch joints give 172.5 degrees, not 180. On a table the pen must be tilted, on a ring round the base; a vertical board works only with the pen pointing away.

Task-constrained workspace (thesis Exp. 1)



IF ASKED

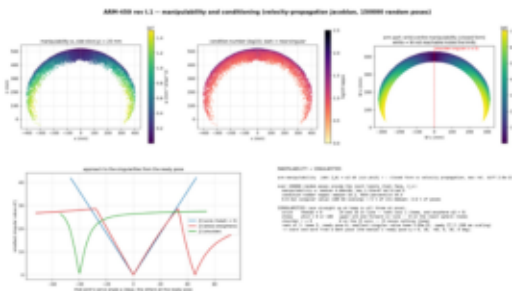
Can that be fixed? -- An angled pen holder (20-30 degrees), or larger joint ranges -- which the parts' clearances set.

Slide 12 -- Manipulability and singularities

SAY

Home, straight up, is singular three ways: shoulder, elbow and wrist. That is why we start from the bent ready pose.

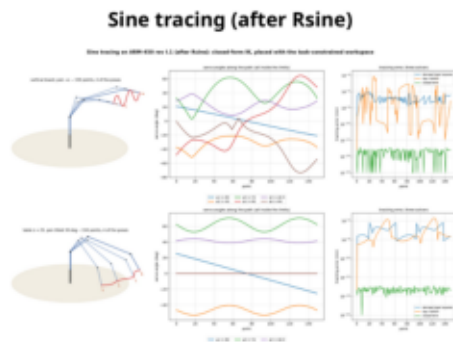
Manipulability and singularities (thesis Exp. 3)



IF ASKED

Where is the arm most dexterous? -- Mid-reach with the elbow bent, like the ring in the myCobot thesis.

Slide 13 -- Sine tracing



SAY

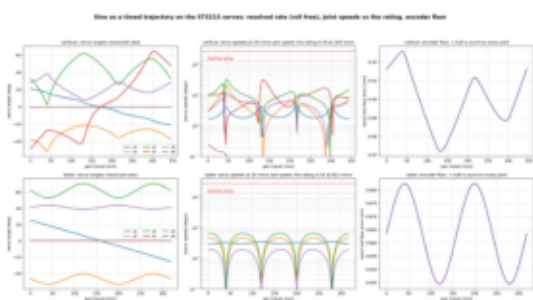
Like Rsine: placed with the task workspace, solved in closed form, re-traced by DLS and by ikpy. A pen is symmetric, so its roll is free: choosing it per point keeps the motion smooth past the wrist singularity.

IF ASKED

How smooth? -- Largest joint step 2.82 deg on the board.

Slide 14 -- Timing on the servos

The sine on the real servos: timing and encoder floor



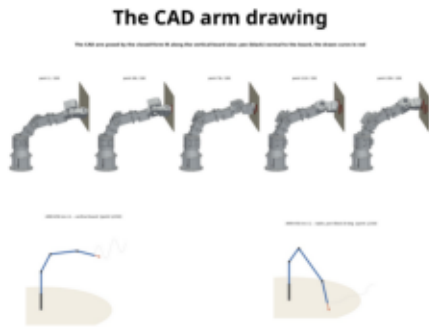
SAY

Timed against the ST3215 rating (270 deg/s): the board sine can run at about 82 mm/s with the servos at half their rating. The encoder alone limits accuracy to about 0.93 mm.

IF ASKED

Why half the rating? -- The rating is no-load; carrying the arm the servos are slower.

Slide 15 -- The CAD arm drawing



SAY

The real CAD posed by the IK, and the animations of both sines.

IF ASKED

Can I see it in RViz? -- Yes: `ros2 launch sine_demo.launch.py` which:=vertical.

Slide 16 -- Conclusions

Conclusions

DH table follows from the CAD with my frame rules: FK = CAD = URDF to 1e-13 mm
Velocity-propagation Jacobian = geometric form; manipulability / conditioning mapped
Spherical wrist -> closed-form IK (exact, all branches); numerical solvers are local; use them along paths
Home is triply singular: start from the bent ready pose
Joint limits cap the pitch at 172.5 deg; no pen straight down; tilted pen on a ring; board from ~190 mm
Free pen roll keeps the traced motion smooth (largest step < 3 deg) -- redundancy, as in the thesis
On the ST3215: board sine at ~80 mm/s with the servos at half their rating; encoder floor ~0.9 mm

SAY

Read the bullets; end on the design lesson: the joint limits, not the link lengths, decide what the arm can draw.

IF ASKED

What next? -- An angled pen holder, then the same trajectory on the printed arm.